

UGB Toppers Mock-2

Maximum Score: 80 • Duration: 120 Minutes

Instructions. Attempt all eight problems. Each problem carries equal weight. Clearly state every claim you use, and justify each step. Calculators and reference materials are not permitted.

Q1

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function satisfying

$$\int_0^1 f(x) dx = 0, \quad \int_0^1 xf(x) dx = 0, \quad \int_0^1 x^2 f(x) dx = 1.$$

- (a) Prove that there exist two distinct points $0 < c_1 < c_2 < 1$ at which f vanishes.
- (b) Prove that there exists a point $c \in (0, 1)$ at which $|f(c)| > 16$.

Q2

Let p be an odd prime. Determine all triples (x, y, n) of positive integers with $n \geq 2$, $\gcd(x, y) = 1$ and

$$x^n + y^n = p^{n-1}(x + y).$$

Prove that your list is complete.

Q3

Let $n \geq 3$ be an integer. For each sequence $\varepsilon = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)$ with each $\varepsilon_i \in \{-1, +1\}$, write

$$M(\varepsilon) = \max_{1 \leq k \leq n} |\varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_k|.$$

- (a) Prove that for every integer k with $1 \leq k \leq n$, the number of sequences $\varepsilon \in \{-1, +1\}^n$ with $M(\varepsilon) \leq k$ equals

$$\sum_{j \in \mathbb{Z}} (-1)^j \binom{n}{\lfloor n/2 \rfloor + j(k+1)}.$$

(The sum has only finitely many nonzero terms.)

- (b) Deduce that at least half of the 2^n sequences ε satisfy $M(\varepsilon) \geq \lfloor \sqrt{n} \rfloor$.

Q4

Let ABC be a scalene triangle with circumcircle Ω and incenter I . Let the line AI meet Ω a second time at M . A circle centered at M passing through B and C meets the line AI at a point S outside the triangle. Prove that S lies on the circle through B, I, C , and that $AS \cdot AI = AB \cdot AC$.

Q5

Let $P(x)$ be a polynomial with real coefficients such that

$$P(x)P(x-1) = P(x^2) \quad \text{for every } x \in \mathbb{R}.$$

- (a) Prove that if $\alpha \in \mathbb{C}$ is a root of P , then $|\alpha| \in \{0, 1\}$, and $|\alpha + 1| \in \{0, 1\}$.
- (b) Hence determine all polynomials P satisfying the given identity.

Q6

Define the sequence $(x_n)_{n \geq 1}$ by

$$x_1 = \frac{1}{2}, \quad x_{n+1} = x_n - x_n^2 \quad \text{for } n \geq 1.$$

- (a) Prove that $0 < x_{n+1} < x_n < 1$ for every $n \geq 1$ and that $x_n \rightarrow 0$ as $n \rightarrow \infty$.
- (b) Prove that $n x_n \rightarrow 1$ as $n \rightarrow \infty$.
- (c) Prove that

$$\lim_{n \rightarrow \infty} \frac{n - n^2 x_n}{\log n} = 1.$$

Q7

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the following three conditions:

- (i) $f(x + y) + f(x - y) = 2f(x)f(y)$ for all $x, y \in \mathbb{R}$;
- (ii) f is bounded on some open interval containing 0;
- (iii) there exists a real number $a \neq 0$ with $f(a) = 1$.

Determine all such functions f , and prove that your list is complete.

Q8

Let a_1, a_2, a_3 be positive reals with $a_1 + a_2 + a_3 = 1$. Prove that

$$\sum_{i=1}^3 \left(\frac{a_i}{1 - a_i} \prod_{j=1}^3 \frac{a_j}{a_i + a_j} \right) \leq \frac{3}{16}.$$

Determine all cases of equality.

End of paper